

Lesson 1.

Net Present Value

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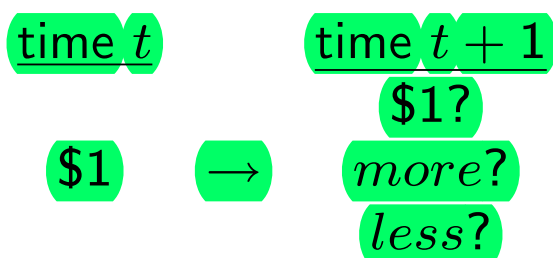
1. Introduction

When deciding to invest or not, a firm or an individual has to decide what to do with the money today.

So we need to compare money today with money in the future.

What's the relationship between \$1 today and \$1 tomorrow?

Is it worth the same \$1 today as \$1 tomorrow/yesterday?

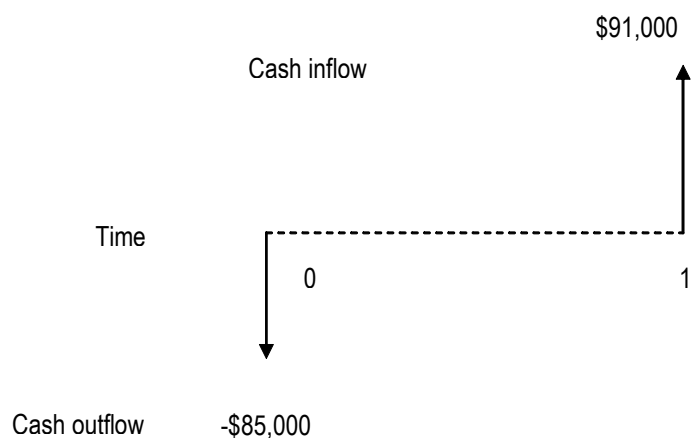


This is what is called the “time-value-of-money” concept.

2. The one-period case

Three equivalent concepts: future value, present value and net present value.

Example (page 61 in book): a financial analyst at a leading real estate firm is thinking about recommending that Kaufman & Broad invest in a piece of land that costs \$85,000. She is certain that next year the land will be worth \$91,000, a sure \$6,000 gain. Given that the guaranteed interest rate in the bank is 10%, should Kaufman & Broad undertake the investment in land?



2.1 Future value (or compound value): the value of a sum after investing over one or more periods.

If the money is invested in the bank, next year they would have

$$\$85,000 \times (1 + 0.1) = \$93,500$$

since

$$\text{future value } \$93,500 > \$91,000$$

then

Invest everything in the bank.

The general formula is

$$FV = C_0 \times (1 + r) \quad (1)$$

where

- C_0 is cash flow today (at time 0),
- and r is the appropriate interest rate.

2.2 Present value: the amount of money that should be put in the alternative investment (the bank) in order to obtain the expected amount next year.

In our example, present value is,

$$PV \times (1 + 0.1) = \$91,000$$

solving for PV

$$PV = \frac{\$91,000}{1.1} = \$82,727.27$$

since

$$\text{present value } \$82,727.27 < \$85,000$$

then

Do not to buy the land.

The general formula is

$$PV = \frac{C_1}{1 + r} \quad (2)$$

where

- C_1 is cash flow at date 1,
- and r is the appropriate interest rate (the one required to buy the land) also called *discount rate*.

2.3 Net present value: the present value of future cash flows minus the present value of the cost of the investment.

The formula is

$$NPV = PV - Cost. \quad (3)$$

In our case, we would have

$$NPV = \frac{\$91,000}{1.1} - \$85,000 = -\$2,273$$

Because

$$NPV < 0$$

Purchase of the land should not be recommended.

NOTE: all three methods reach the same conclusions.

3. The multiperiod case

3.1 Future value and compounding

General formula

$$FV = C_0 \times (1 + r)^T \quad (4)$$

where

- C_0 is cash flow at date 0,
- r is the appropriate interest rate, and
- T is the number of periods over which the cash is invested.

Example: \$1.10 of dividend, expected to grow at a 40% per year over the next 2 years, then in 2 years the dividend will be

$$FV = C_0 \times (1 + r)^T = \$1.10 \times (1.40)^2 = \$2.156$$

Notice that

$$(1 + 0.40)^2 = 1 + 0.40^2 + 2 \times 0.40 \quad (5)$$

Compounding: the process of leaving money in the capital market and lending it for another year.



Notice that

$$\$2.156 > \$1.10 + 2 \times [\$1.10 \times 0.40] = \$1.98$$

Simple interest: interest payments are not reinvested, i.e. $T \times r$. In our example,

$$2 \times 0.40 = 0.80$$

Compound interest: each interest payment is reinvested, i.e. r^T . In our example,

$$0.40^2 = 0.16$$

then

$$\$2.156 = \$1.10 \times (1 + 0.16 + 0.80)$$

What rate is enough? Alternatively, we may not know the interest rate. Example:

Assume the total cost of a college education will be \$50,000 when your child enters college in 12 years. You have \$5,000 to invest today. What rate of interest must you earn on your investment to cover the cost of your child's education?

$$FV = C_0(1 + r)^T$$

$$\$50,000 = \$5,000(1 + r)^{12}$$

$$\frac{\$50,000}{\$5,000} = (1 + r)^{12}$$

$$r = \left(\frac{50,000}{5,000} \right)^{\frac{1}{12}} - 1 = 0.2115$$

3.2 Present value and compounding

Discounting: the process of calculating the present value of a future cash flow. It is the opposite of compounding.

Example: how much would an investor have to set aside today in order to have \$20,000 five years from now if the current rate is 15%?

$$PV(1 + r)^T = C_T \rightarrow PV = \frac{C_T}{(1 + r)^T}$$

$$PV(1 + 0.15)^5 = \$20,000$$

$$PV = \frac{\$20,000}{(1.15)^5} = \$9,943.53$$

Present value factor: the factor used to calculate the present value of a future cash flow.

$$\frac{1}{(1 + r)^T}$$

3.3 Multiperiod net present value

NPV of a T —period project can be written as



$$NPV = -C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_T}{(1+r)^T}$$

$$NPV = -C_0 + \sum_{i=1}^T \frac{C_i}{(1+r)^i}. \quad (6)$$

4. Compounding periods

Compounding an investment m times a year for T years provides for future value of wealth

$$FV = C_0 \left(1 + \frac{r}{m} \right)^{mT} \quad (7)$$

where

- C_0 is the initial investment,
- r is the stated annual interest rate (also called *annual percentage rate*),
- m is how many times a year the investment is compounded, and
- T is the total number of periods of the investment.

Example: if you invest \$50 for 3 years at 12% compounded semiannually, your investment will grow to

$$FV = \$50 \left(1 + \frac{0.12}{2}\right)^{2 \times 3} = \$50(1 + 0.06)^6 = \$70.93$$

Example: what is the end-of-year wealth of \$1 invested for one year at a stated annual interest rate of 24% compounded monthly?

$$FV = \$1 \left(1 + \frac{0.24}{12}\right)^{12 \times 1} = \$1(1 + 0.02)^{12} = \$1.2682$$

Example: what is the wealth at the end of 5 years of investing \$5,000 at a stated annual interest rate of 12% per year, compounded quarterly?

$$FV = \$5,000 \left(1 + \frac{0.12}{4}\right)^{4 \times 5} =$$
$$\$5,000(1 + 0.03)^{20} = \$9,030.50$$

Effective annual interest rates: the annual rate that would give us the same end-of-investment wealth after T years.

In the example above,

$$\$50(1 + EAR)^3 = \$70.93$$

$$EAR = \left(\frac{70.93}{50}\right)^{\frac{1}{3}} - 1 = 0.1236 = 12.36\%.$$

So, investing at 12.36% compounded annually is the same as investing at 12% compounded semiannually.

That is,

$$(1 + r_{EAR}) = \left(1 + \frac{r}{m}\right)^m \quad (8)$$

$$r_{EAR} = \left(1 + \frac{r}{m}\right)^m - 1. \quad (9)$$

Continuous Compounding

The general formula for the future value of an investment compounded continuously over many periods can be written as

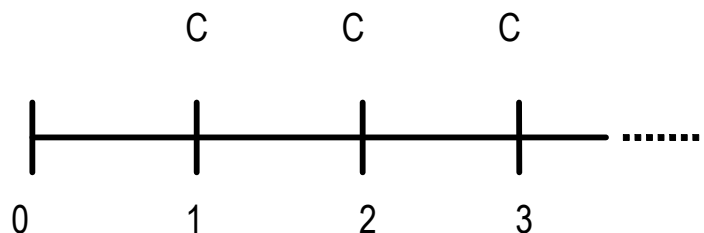
$$FV = C_0 \times e^{rT} = C_0 \times \exp(rT) \quad (10)$$

where

- C_0 is cash flow at date 0,
- r is the stated annual interest rate,
- T is the number of periods over which the cash is invested.

5. Simplifications

Perpetuity: a constant stream of cash flows that lasts forever.



$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots = \frac{C}{r}. \quad (11)$$

Example: the British bonds called *consols*. The present value of a consol is the present value of all of its future coupons.

Example: consider a perpetuity paying \$100 a year. If the relevant interest rate is 8%, what is the value of the consol?

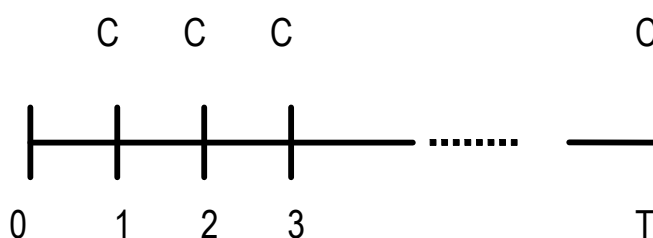
$$PV = \frac{\$100}{0.08} = \$1,250$$

What is the value of the consol if the interest rate goes down to 6%?

$$PV = \frac{\$100}{0.06} = \$1,666.67$$

Note: the value of the perpetuity rises with a drop in the interest rate, and vice versa.

Annuity: a stream of constant cash flows that lasts for a fixed number of periods.



$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots + \frac{C}{(1+r)^T} \quad (12)$$

$$PV = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right] \quad (13)$$

Intuition: an annuity is valued as the difference between two perpetuities: one that starts at time 1 minus another one that starts at time $T + 1$.

Example: If you can afford a \$400 monthly car payment, how much car can you afford if interest rates are 7% on 36 – *month* loans?

$$PV = \frac{\$400}{\frac{0.07}{12}} \left[1 - \frac{1}{\left(1 + \frac{0.07}{12}\right)^{36}} \right] = \$12,954.59$$

Example: What is the present value of a 4–*year* annuity of \$1000 per year that makes its first payment 2 years from today if the discount rate is 9%?

First, let's bring all the value to the starting time of the annuity

$$PV = \$1000 + \frac{\$1000}{1.09} + \frac{\$1000}{1.09^2} + \frac{\$1000}{1.09^3} = \$3531.29$$

this amount today is

$$PV = \frac{\$3531.29}{1.09^2} = \$2972.22$$

6. Net present value: first principles of finance

Making consumption choices over time: the intertemporal budget constraint

$$PV(\text{consumption}) \leq PV(\text{total wealth})$$

So, given

- Initial assets A_0
- Annual wage w_t at any period t
- Real interest rate r
- Living from $t = 0$ to $t = T$

and individual's consumption should be such that

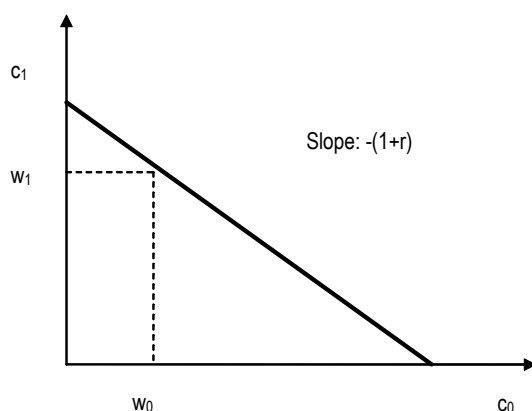
$$c_0 + \frac{c_1}{1+r} + \dots + \frac{c_T}{(1+r)^T} \leq A_0 + w_0 + \frac{w_1}{1+r} + \dots + \frac{w_T}{(1+r)^T}$$

Let's consider a person who lives for two periods. This person's budget constraint can be explained with the net present value.

Assumptions:

- no initial assets,
- income w_0 and w_1 ,
- r is the real interest rate in the financial market.

This can be represented



$$c_0 + \frac{c_1}{1+r} \leq w_0 + \frac{w_1}{1+r}$$

if all is consumed today (i.e. $c_1 = 0$), then

$$c_0 = w_0 + \frac{w_1}{1+r}$$

but if all is saved and consumed tomorrow (i.e. $c_0 = 0$), then

$$c_1 = w_0(1+r) + w_1,$$

which are the intercepts to plot the budget constraint.

Changes in the interest rate change the slope of the budget constraint.

Changes in income shift the budget constraint.

Making consumption choices over time: consumer preferences Individual's preferences can be represented by utility functions.

Indifference curves (IC): combinations of goods that give the consumer the same level of happiness or satisfaction.

Some properties:

- higher IC are preferred to lower IC
- IC are downward sloping (the slope is the *marginal rate of substitution*)

- IC do not cross each other
- IC are bowed inward (convex)

Making consumption choices over time: optimal consumption choice

Individuals choose consumption intertemporally by maximizing their utility subject to the budget constraint.

For our example, the problem can be written as follows:

$$\max_{c_0, c_1} u(c_0, c_1)$$

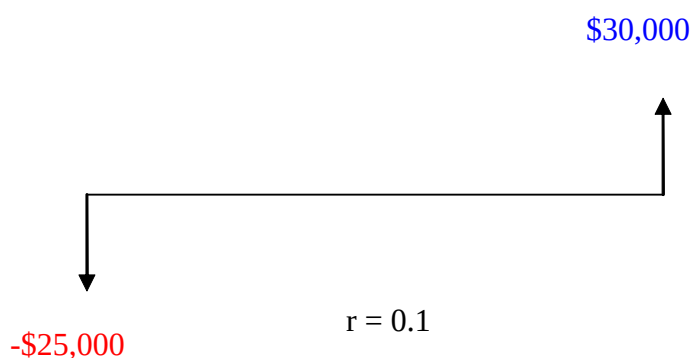
subject to

$$c_0 + \frac{c_1}{1+r} \leq w_0 + \frac{w_1}{1+r}.$$

Illustrating the investment decision

Consider an investor who has an initial endowment of income, w_0 , of \$40,000 this year and $w_1 = \$55,000$ next year.

Suppose that he faces a 10% interest rate and is offered the following investment:



First of all, should the individual undertake the project?
Compute NPV

$$NPV = -\$25,000 + \frac{\$30,000}{1.1} = \$2,272.72 > 0 \rightarrow \text{YES!}$$

Budget constraint without investment

$$c_1 + c_0(1 + r) = w_0(1 + r) + w_1,$$

that is

$$c_1 + 1.1c_0 = \$99,000$$

Budget constraint with investment Notice that if he invests at $t = 0$ he will have left

$$w_0 - \$25,000 - c_0.$$

At time $t = 1$ he will get

$$w_1 + \$30,000.$$

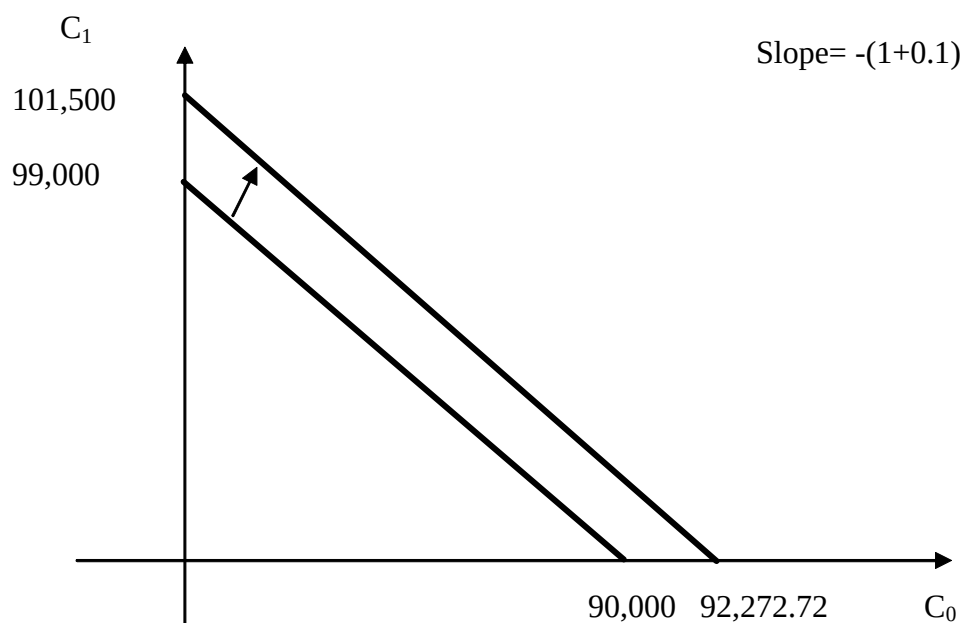
Then the budget constraint becomes

$$\begin{aligned} c_1 + c_0(1.1) &= \\ &= (\$40,000 - \$25,000)(1.1) + \$55,000 + \$30,000 \end{aligned}$$

that is

$$c_1 + 1.1c_0 = \$101,500$$

If we plot both budget constraints we obtain



That is, the consumption possibilities frontier is expanded if we undertake this investment project, which has a $\text{NPV} > 0$.

Where will the consumer stay?

We need some assumption on the preferences.

Assume $u(c_0, c_1) = c_0c_1$. Then the problem to solve is

$$\max_{c_0, c_1} c_0c_1$$

subject to

$$c_1 + 1.1c_0 = \$99,000$$

in the case of no investment.

Optimal solution (A) is:

$$(c_0^*, c_1^*) = (\$45,000, \$49,500).$$

If we consider the investment opportunity, the problem to solve is

$$\max_{c_0, c_1} c_0c_1$$

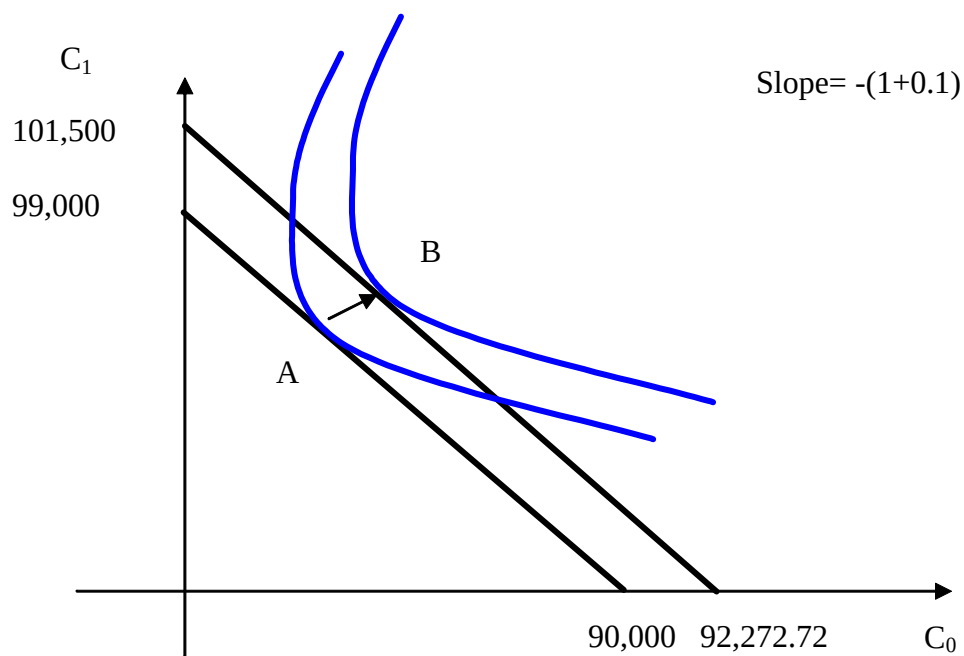
subject to

$$c_1 + 1.1c_0 = \$101,500$$

and the optimal solution (B) is:

$$(c_0^*, c_1^*) = (\$46,136.36, \$50,749.99).$$

If we plot it, we have



The investment decision allows the individual to reach a higher indifference curve, and therefore, he is better off.